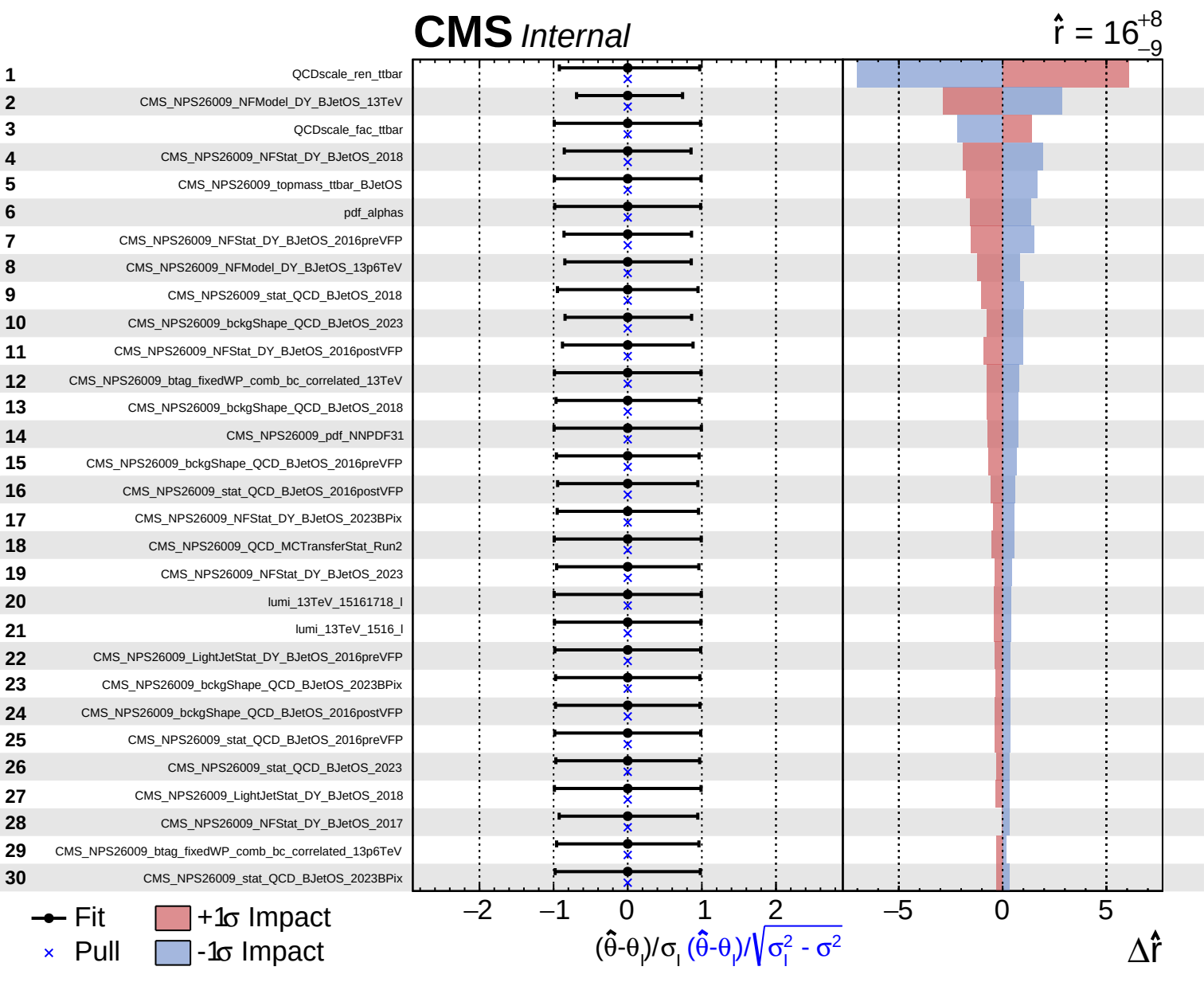
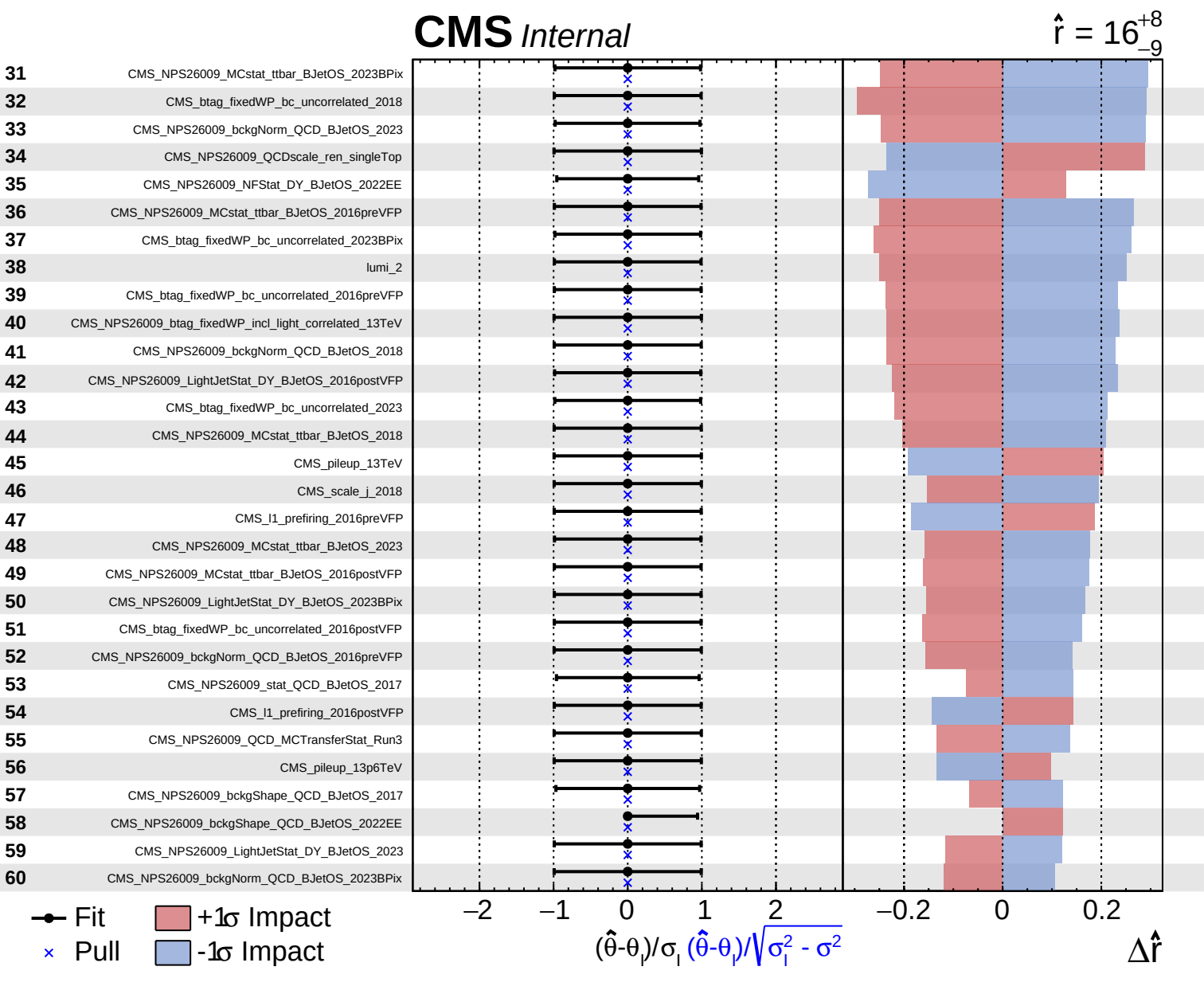
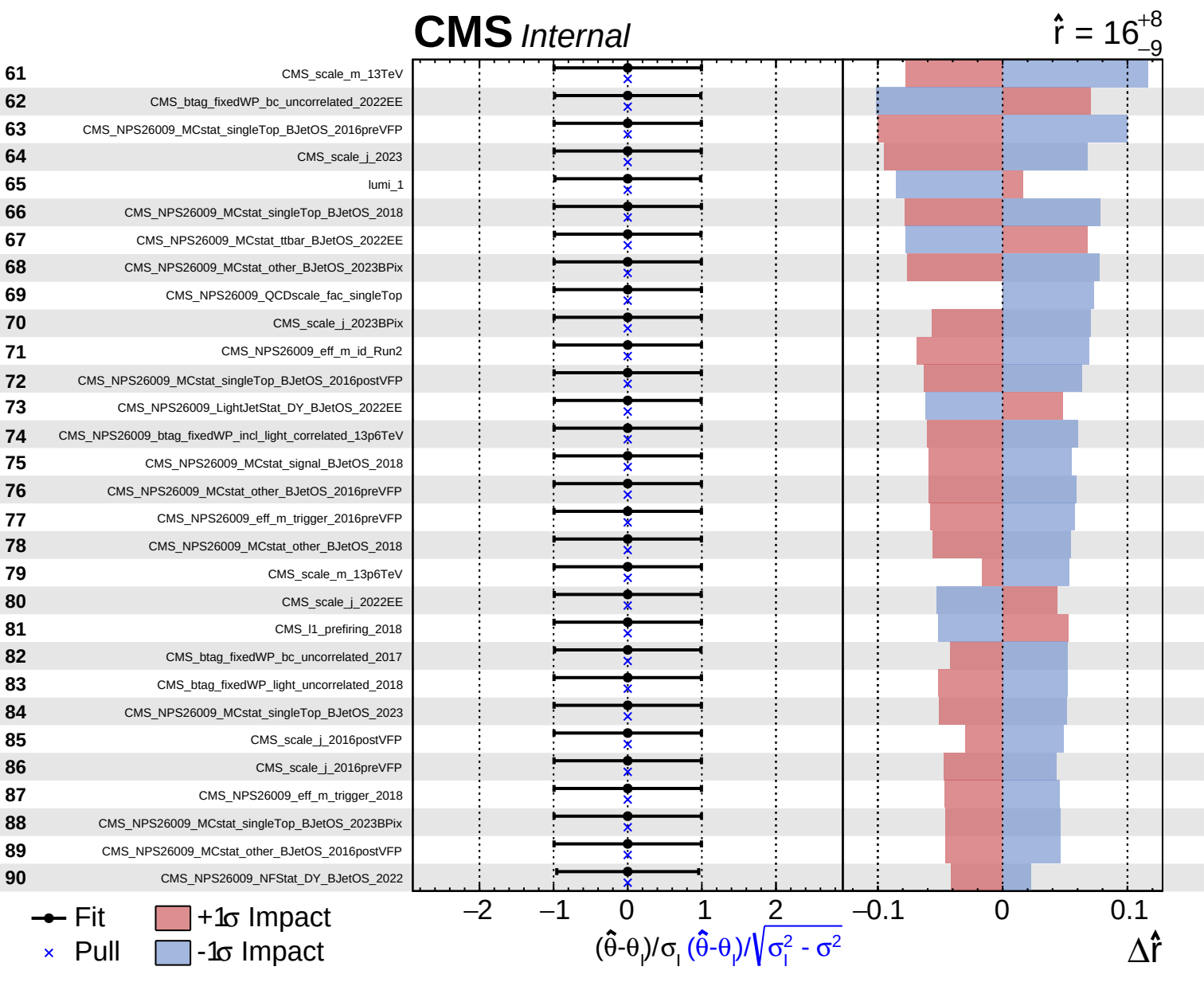
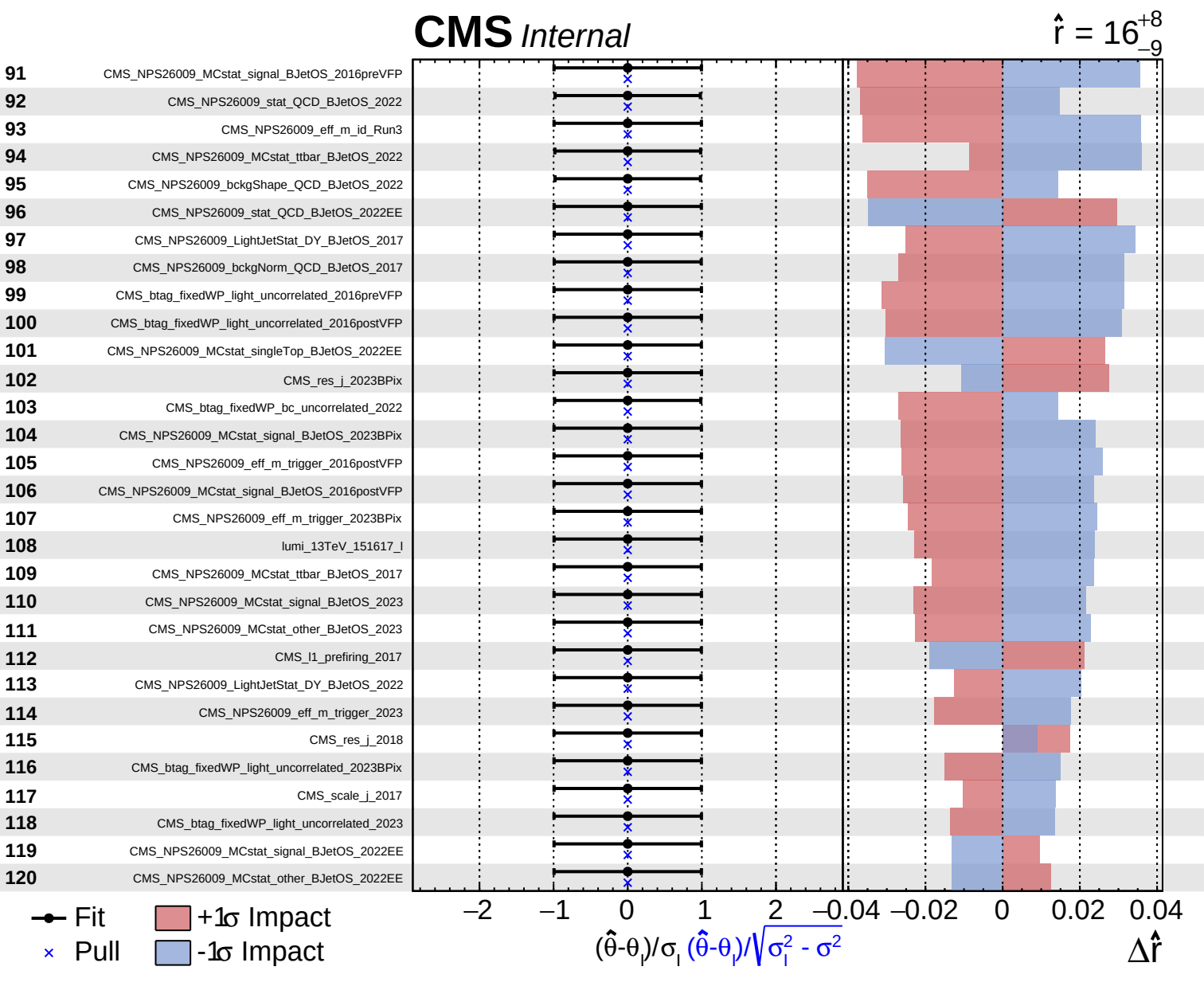
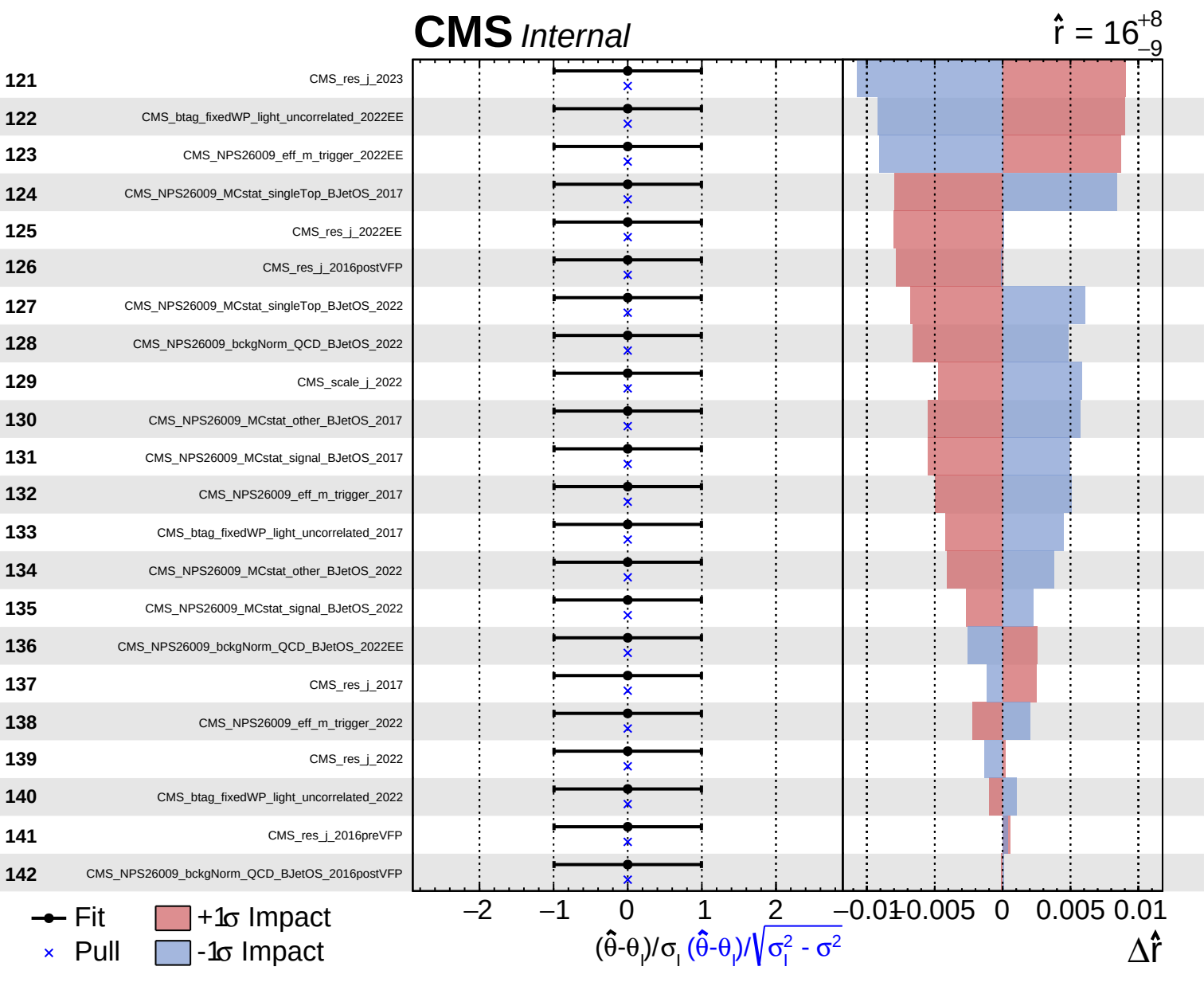

CMS Internal
$\hat{r} = 16^{+8}_{-9}$
1 QCDscale_ren_ttbar
2 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
3 QCDscale_fac_ttbar
4 CMS_NPS26009_NFStat_DY_BJetOS_2018
5 CMS_NPS26009_topmass_ttbar_BJetOS
6 pdf_alphas
7 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
8 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
9 CMS_NPS26009_stat_QCD_BJetOS_2018
10 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
11 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
12 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
13 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
14 CMS_NPS26009_pdf_NNPDF31
15 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
16 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
17 CMS_NPS26009_NFStat_DY_BJetOS_2023BPix
18 CMS_NPS26009_QCD_MCTransferStat_Run2
19 CMS_NPS26009_NFStat_DY_BJetOS_2023
20 lumi_13TeV_15161718_l
21 lumi_13TeV_1516_l
22 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
23 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
24 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
25 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
26 CMS_NPS26009_stat_QCD_BJetOS_2023
27 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
28 CMS_NPS26009_NFStat_DY_BJetOS_2017
29 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
30 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
−2 −1 0 1 2
−5 0 5
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 16^{+8}_{-9}$
31 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
32 CMS_btag_fixedWP_bc_uncorrelated_2018
33 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
34 CMS_NPS26009_QCDscale_ren_singleTop
35 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
36 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
37 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
38 lumi_2
39 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
40 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
41 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
42 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
43 CMS_btag_fixedWP_bc_uncorrelated_2023
44 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
45 CMS_pileup_13TeV
46 CMS_scale_j_2018
47 CMS_l1_prefiring_2016preVFP
48 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
49 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
50 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
51 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
52 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
53 CMS_NPS26009_stat_QCD_BJetOS_2017
54 CMS_l1_prefiring_2016postVFP
55 CMS_NPS26009_QCD_MCTransferStat_Run3
56 CMS_pileup_13p6TeV
57 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
58 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
59 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
60 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
−2
−1
0
1
2
−0.2
0
0.2
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 16^{+8}_{-9}$
61 CMS_scale_m_13TeV
62 CMS_btag_fixedWP_bc_uncorrelated_2022EE
63 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
64 CMS_scale_j_2023
65 lumi_1
66 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
67 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
68 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
69 CMS_NPS26009_QCDscale_fac_singleTop
70 CMS_scale_j_2023BPix
71 CMS_NPS26009_eff_m_id_Run2
72 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
73 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
74 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
75 CMS_NPS26009_MCstat_signal_BJetOS_2018
76 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
77 CMS_NPS26009_eff_m_trigger_2016preVFP
78 CMS_NPS26009_MCstat_other_BJetOS_2018
79 CMS_scale_m_13p6TeV
80 CMS_scale_j_2022EE
81 CMS_l1_prefiring_2018
82 CMS_btag_fixedWP_bc_uncorrelated_2017
83 CMS_btag_fixedWP_light_uncorrelated_2018
84 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
85 CMS_scale_j_2016postVFP
86 CMS_scale_j_2016preVFP
87 CMS_NPS26009_eff_m_trigger_2018
88 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
89 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
90 CMS_NPS26009_NFStat_DY_BJetOS_2022
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.1
0
0.1
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 16^{+8}_{-9}$
91 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
92 CMS_NPS26009_stat_QCD_BJetOS_2022
93 CMS_NPS26009_eff_m_id_Run3
94 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
95 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
96 CMS_NPS26009_stat_QCD_BJetOS_2022EE
97 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
98 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
99 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
100 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
101 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
102 CMS_res_j_2023BPix
103 CMS_btag_fixedWP_bc_uncorrelated_2022
104 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
105 CMS_NPS26009_eff_m_trigger_2016postVFP
106 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
107 CMS_NPS26009_eff_m_trigger_2023BPix
108 lumi_13TeV_151617_l
109 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
110 CMS_NPS26009_MCstat_signal_BJetOS_2023
111 CMS_NPS26009_MCstat_other_BJetOS_2023
112 CMS_l1_prefiring_2017
113 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
114 CMS_NPS26009_eff_m_trigger_2023
115 CMS_res_j_2018
116 CMS_btag_fixedWP_light_uncorrelated_2023BPix
117 CMS_scale_j_2017
118 CMS_btag_fixedWP_light_uncorrelated_2023
119 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
120 CMS_NPS26009_MCstat_other_BJetOS_2022EE
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.04
−0.02
0
0.02
0.04
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 16^{+8}_{-9}$
121 CMS_res_j_2023
122 CMS_btag_fixedWP_light_uncorrelated_2022EE
123 CMS_NPS26009_eff_m_trigger_2022EE
124 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
125 CMS_res_j_2022EE
126 CMS_res_j_2016postVFP
127 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
128 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
129 CMS_scale_j_2022
130 CMS_NPS26009_MCstat_other_BJetOS_2017
131 CMS_NPS26009_MCstat_signal_BJetOS_2017
132 CMS_NPS26009_eff_m_trigger_2017
133 CMS_btag_fixedWP_light_uncorrelated_2017
134 CMS_NPS26009_MCstat_other_BJetOS_2022
135 CMS_NPS26009_MCstat_signal_BJetOS_2022
136 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
137 CMS_res_j_2017
138 CMS_NPS26009_eff_m_trigger_2022
139 CMS_res_j_2022
140 CMS_btag_fixedWP_light_uncorrelated_2022
141 CMS_res_j_2016preVFP
142 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.01 −0.005 0 0.005 0.01
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$